Noise-tailored two-qubit gates on a high-contrast featured dephasing model

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This note is the companion to the June 11 report: there, the paper's pipeline ran on the experimentally anchored Tarucha-class model and the noise-tailoring margin came out a reproducible but modest 1.5–1.7×. *Here the same pipeline runs on a* featured *model — every structural ingredient still a measured type of spectral feature, but their placement and contrast chosen so that the value of the full spectral reconstruction is visible at a glance. Three requirements were imposed in the June 12 discussion: the working infidelities must be realistic for today's devices; the reconstruction must genuinely capture the model — every harmonic of all six spectra within its stated error bars; and the demonstration must articulate what the two-qubit part of the reconstruction buys. Throughout, the optimizer never sees ground truth, and every quoted "true" infidelity is evaluated exactly on the analytic model. All numbers June 12, 2026: the characterization is one 128k-shot reconstruction (everything downstream — designs, budgets, bands — is built on it), and the SPAM comparison is three 64k arms run on identical noise.*

1 The noise model: what we assume, and why

Two correlated local fields, not one shared bath. The skeleton is unchanged from the anchored model: two partially correlated local electric fields e_A, e_B (shared fraction 0.8), the exchange coupling responding to their *difference*, $\zeta_{12} = a\,e_A - b\,e_B$ with $b \gtrsim a$, plus independent qubit-local processes. This is the minimal structure behind Yoneda's measured $(+,+,-)$ coherence sign pattern, and it pins the signs and relative phases of all three cross-spectra rather than leaving them free. Four ingredients are then recomposed on this skeleton, each a measured feature type with its amplitude as the stated modeling choice.

A slow, common-mode carrier holds the coherence budget. A slow bath (correlation time $\sim 400\,\tau$, in-band ω^{-2} tail) carries the entire $T_2^* = 3500\,\tau$ per qubit — the measured Ge-hole value (17.6 μs at the $\tau = 5$ ns anchor), equivalently purified ^{28}Si at $\tau \approx 7$ ns. Its in-band tail is what defeats the low CDD orders. Structurally, the carrier is *common-mode* between the qubits at fraction $c_{\mathrm{QS}} = 0.95$, with the remainder qubit-local: slow drifts of this class (global field, thermal, common charge motion) hit both qubits together, and inter-qubit coherences up to 0.7 are what Yoneda measured at low frequency. Two consequences are wired into the construction: the carrier appears in $\mathrm{Re}\,S_{1,2}$ (and only there — a common-mode drift cancels in the difference coupling that drives ζ_{12}, so $S_{12,12}$, $S_{1,12}$, $S_{2,12}$ carry none of it), and its *sign* at low frequency is a measurable prediction with design consequences (Sec. 3).

The trap structure: a defect line with harmonics. On the qubit-local channels, a narrow defect line at $\omega_0\tau = 0.051$ with harmonics at $\{2, 3, 4\}\,\omega_0$ (cf. the ^{73}Ge Larmor line and its measured second harmonic surviving isotopic purification), plus one line at $\omega\tau = 0.372$ covering the top of the band reachable under the control-bandwidth constraint below. The placement is the engineered part: the CDD library's passbands at $T_G = 320\tau$ fall on the first, second, and fourth harmonic for orders 3/4/5, and because the passbands scale as $1/T_G$ while the lines are fixed, *every doubling of the gate time slides the library down onto the next line.* This is why the margins in Sec. 4 grow with gate time. The lines are strictly *local* defect structure: they appear on $S_{1,1}$ and $S_{2,2}$ only, and the inter-qubit coherence dips at the line frequencies.

A line on the exchange channel only. An independent coupler-defect process (TLF resonance at $\omega\tau = 0.236$ with a weak Connors-type knee) enters ζ_{12} alone. Single-qubit QNS cannot see it; it is structure reserved for the two-qubit spectra, and it is what the idle gate's ZZ-decoupling has to dodge.

A quiet — but measurable — electrical background. A $1/f^{0.9}$ floor (the measured charge-noise exponent class) roughly 25 dB below the anchored natural-Si level. This is the contrast knob, and it is deliberately placed *at* the reconstruction's sensitivity rather than below it: between the fourth harmonic and the 0.372 line it leaves the quiet band $\omega\tau \in [0.26, 0.31]$ whose *measured* level, not any line, sets what an informed design can reach.

The scenario parameter. A minimum pulse separation of 8τ (40 ns π-pulse spacing at the anchor) applies identically to the CDD library, the NT optimizer, and every counterfactual. Without it, separation-limited trains park their passbands above the comb's measured band and the comparison decouples from the reconstruction; with it, every sequence lives on spectra the protocol has measured.

quantity	value	anchor
time unit (min. pulse separation /8)	$\tau = 5$ ns	Ge-hole / fast-Rabi anchor
T_2^* (both qubits)	$3500\,\tau = 17.6\,\mu$s	measured Ge-hole T_2^*
Gaussian FID $T_2 = 2/S_{a,a}(0)$	$3.10 \times 10^3\,\tau$	model output
slow-carrier correlation time / share	$400\,\tau$ / $c_{\mathrm{QS}} = 0.95$ common-mode	Yoneda low-f coherence class
defect line + harmonics	$\omega_0\tau = 0.051$, $\{2,3,4\}\omega_0$, 0.372	^{73}Ge-line / trap family
coupler (ZZ) line + knee	$\omega\tau = 0.236$; knee at 0.025	shared-TLF class, stylized
electrical floor	$1/f^{0.9}$, -25 dB vs anchored	charge-noise exponent class
quiet band (between lines)	$\omega\tau \in [0.26, 0.31]$	set by the engineered lines
control bandwidth	min. pulse separation $8\tau = 40$ ns	scenario parameter
gate windows studied	CZ: $T_G = 40$–$2560\,\tau$; idle: 320–$10240\,\tau$	0.01–$3.3\,T_2^*$

Table 1: The dephasing scale and where the gates sit relative to it.

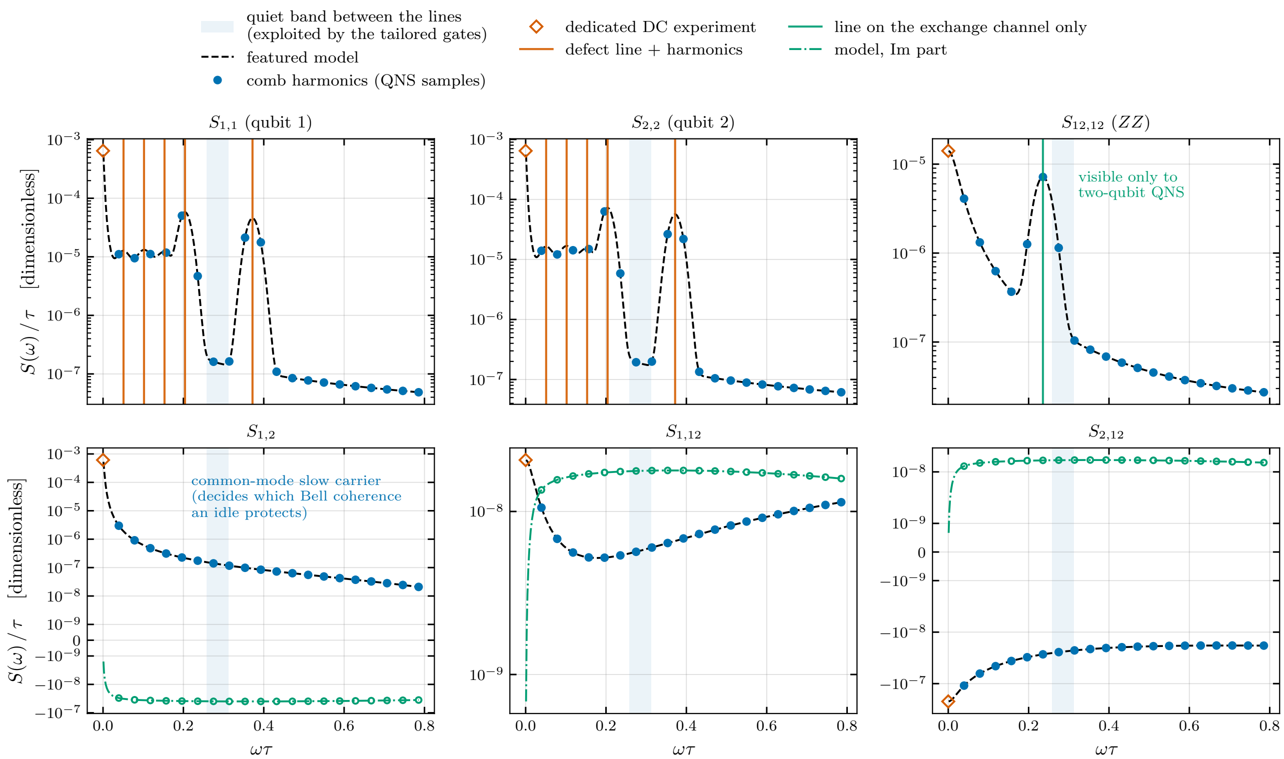

Figure 1: The featured two-qubit dephasing model. Each panel is one of the six spectra the QNS protocol reconstructs: dashed black is the analytic model (top row: the two single-qubit channels and the ZZ channel; bottom row: the three cross-spectra, real parts in black, imaginary parts in green). Blue dots mark the 20 comb harmonics $\omega_k\tau = 2\pi k/160$ the protocol samples; open diamonds are the dedicated DC experiments. Vertical orange lines: the defect line and its harmonics, which live on the qubit-local channels only; the coupler resonance (green) lives on $S_{12,12}$ *only* — structure invisible to single-qubit QNS. The common-mode slow carrier dominates $\mathrm{Re}\,S_{1,2}$ at low frequency (its low-f rise in the $S_{1,2}$ panel); which Bell coherence an idle protects depends on its measured sign. Shaded band: the quiet window between the lines that the tailored designs exploit.

2 Characterization: the blind reconstruction recovers the truth

The protocol is identical to the anchored study — the same comb, the same estimators, the same DC experiments — run once at the budget a one-time device characterization affords: 128k shots per estimator, with the harmonic sweeps extended to 2560τ ($0.73\,T_2^*$; about fifteen minutes of simulation wall time). The acceptance standard is the one set in the June 12 discussion: *within error bars everywhere, with bars that inform.* Table 2 gives the result against the analytic truth, per spectrum.

spectrum	median rel. dev.	median pull	max pull	within 2σ
$S_{1,1}$	1.8%	0.35	1.80	100%
$S_{2,2}$	0.8%	0.34	2.07	95%
$S_{12,12}$	5.1%	0.79	2.20	85%
$S_{1,2}$	6.6%	0.43	1.96	100%
$S_{1,12}$	16.1%	0.67	2.17	95%
$S_{2,12}$	6.6%	0.54	1.81	100%

Table 2: Blind 128k reconstruction vs. truth (pull = deviation / quoted total uncertainty; 20 harmonics per channel). Every tooth of every channel sits within 2.2σ of truth — across the 120 teeth, the handful in the 2–3σ band is what correct coverage predicts. The bars inform: median signal-to-bar ratios run 4–25 per channel, and every engineered feature (the defect harmonics on $S_{1,1}/S_{2,2}$, the coupler line on $S_{12,12}$, the common-mode carrier in $\mathrm{Re}\,S_{1,2}$) is detected at 5σ or better. The double-echo DC experiment lands $S_{12,12}(0) = 6.19 \times 10^{-7}$ against a true 6.07×10^{-7}.

Two honest footnotes to "tight everywhere". First, harmonics adjacent to the lines on the smallest channels carry locally widened statistical bars: the probe sequences decohere faster while sweeping across a line, a measurement back-action that no shot budget removes; those teeth remain within bars. Second, the slope-fit DC experiments on $S_{1,1}/S_{2,2}$ cannot see the full quasistatic carrier and are flagged *undetermined* by the protocol itself (the stored values are first-harmonic floors); the double-echo $S_{12,12}(0)$ experiment, by contrast, is determined and lands within 2% of truth. The flags propagate to the gate stage, which treats those DC points accordingly — part of the protocol, not a patch.

SPAM. The three-arm comparison (Fig. 2) reruns the full experiment suite with hardware-realistic SPAM injected ($\alpha_M \sim 0.95$–0.97, $\alpha^z_{SP} \sim 0.98$–0.99, nonzero readout asymmetries and transverse preparation errors), at 64k shots per arm on *identical noise* (record/replay), so the arms differ only in what SPAM does and what mitigation recovers. The result reproduces the anchored study's conclusion on the featured landscape: the mitigated arm is statistically indistinguishable from the SPAM-free reference (maximum deviation 0.02σ across all six spectra), while the raw arm is biased at up to 3.2σ per channel — modest in σ's only because this quiet landscape's bars are wide. In *relative* terms the corruption is exactly where a tailored design leans: up to 86–106% on the self-spectra (the carrier head at the first harmonic, a line shoulder), and up to $\sim 9\times$ on the quiet ZZ teeth whose measured level sets the ZZ error floor. The SPAM-robust protocol remains immune by construction at its stated price (it cannot reconstruct $S_{1,12}, S_{2,12}$); whether that price matters is a gate-level question, answered next.

3 What the gate design needs from the spectra, and what the physics does

With a reconstruction that captures the landscape, the question is how much of it the control design actually consumes — and what the parts the design does *not* consume are for. We answer with the same two experiments as the anchored study, run for both gates (Fig. 3), plus the demonstration the cross-spectra exist for (Fig. 4).

Two-qubit QNS is necessary to design the entangler. Re-running the blind NT optimization with the optimizer's channels progressively restored — the two single-qubit spectra alone → the three self-spectra → the SPAM-robust four → all six — the entangling gate steps once, by $3.1\times$, and only at the ZZ rung: 2.28×10^{-3} from the single-qubit spectra alone versus 7.31×10^{-4} once $S_{12,12}$ is supplied, on the true noise at $T_G = 320\tau$ (Fig. 3a). Blind to the ZZ-correlated dephasing the gate accumulates, the single-qubit design drives its own

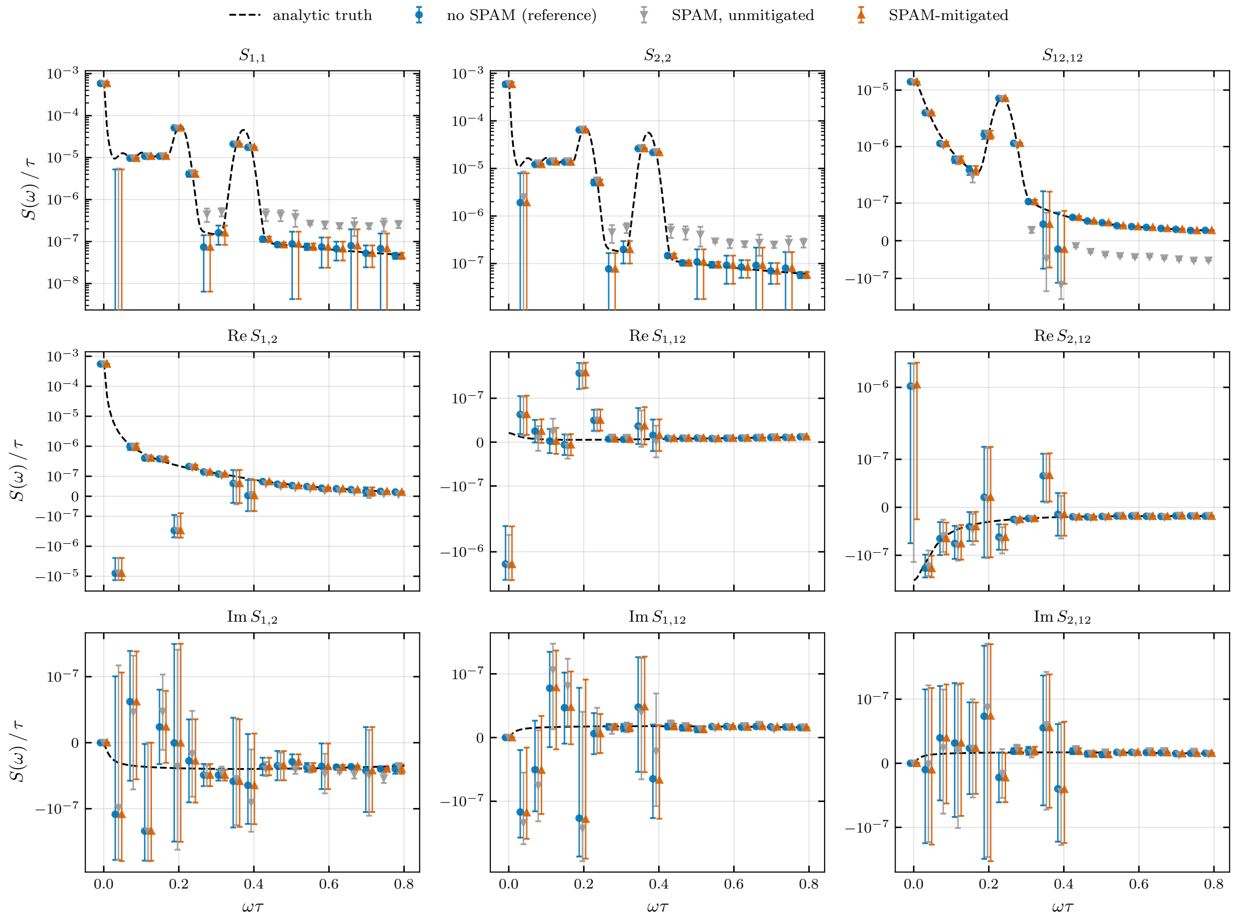

Figure 2: The three reconstruction arms at 64k shots on identical noise, against analytic truth (dashed): reference (no SPAM, blue circles), raw (SPAM, unmitigated, grey triangles), SPAM-mitigated (orange). Bars are the quoted total (statistical + systematic) uncertainties; arms are slightly offset in ω for legibility. Top row: the three self-spectra; middle/bottom rows: real and imaginary parts of the cross-spectra. The raw arm's bias is largest on the line heights; the mitigated arm sits on the reference everywhere.

self-only objective to $\sim 5 \times 10^{-4}$ while sitting at 2.28×10^{-3} in truth; given $S_{12,12}$ the optimizer shapes the y_{12} filter against it. That a two-qubit channel can decide a high-fidelity design is fixed by how the average fidelity reads the noise: to leading order in a dephasing model,

$$1 - F_{\text{pro}} \;=\; \sum_a w_a\, \chi_{aa} \;+\; O(\chi^2), \qquad \chi_{ab} \;=\; \int \frac{d\omega}{2\pi}\, \mathcal{F}_{ab}(\omega)\, S_{ab}(\omega), \tag{1}$$

where $a, b \in \{1, 2, 12\}$ index the Z_1, Z_2, Z_1Z_2 dephasing generators and $w_a > 0$. The three *self*-spectra $(S_{1,1}, S_{2,2}, S_{12,12})$ enter at *first* order; every *cross*-spectrum cancels in the Pauli trace that defines F_{pro} — $\sum_P s_P^a s_P^b = 0$ for $a \neq b$, the $\pm$ filter combinations pairing off — and survives only at *second* order, $O(\chi^2)$. A high-fidelity design is therefore set by whichever self-spectrum the gate cannot escape, here $S_{12,12}$; the cross-spectra (robust-4, all six) add nothing, so the CZ ladder is the single step the first-order argument demands. The idle shows the same step, smaller ($1.19\times$ at 640τ, $1.54\times$ at 10240τ, best over the repetition number M). $S_{12,12}$ — the ZZ-correlated dephasing — is reconstructible only by the two-qubit protocol, and this design gap is its operational necessity, not a characterization nicety.

SPAM-biased spectra. Handing the optimizer the raw arm's biased reconstruction yields the *identical winner at the identical true infidelity* for both gates — the same NT(27,30) entangling design at 7.31×10^{-4} and the same idle winner at 1.57×10^{-3} as the reference and mitigated arms (Fig. 3, right column). On this landscape even the predictions agree across arms (within 0.5%, each sitting $\sim 1.5\times$ above its gate's true value — the uniform conservatism discussed in Sec. 4): the raw arm's corruption is concentrated on line heights and quiet ZZ teeth, frequencies where the *winning* filters carry little weight. That is landscape-dependent luck, not a property of unmitigated data — on the anchored model the raw arm mispredicted where the reference predicted to 0.7% — and it leaves SPAM's measured cost here exactly where Fig. 2 shows it: in the spectroscopy. A device map drawn from raw data mis-states the line heights by up to $\sim 2\times$ and the quiet ZZ floor by $\sim 9\times$; mitigation repairs all of it for two orders of magnitude less than the cost of being wrong about a trap's depth.

What full, unbiased characterization buys is everything around the design:

- **The error budget.** Under the best decoupling the two-qubit channels ($S_{12,12}$ plus cross-spectra) carry **45%** of the entangling gate's residual at $T_G = 320\tau$ and **22.5%** of the deep idle's (2560τ): nearly half of the tailored CZ's error is ZZ-correlated dephasing a single-qubit campaign cannot see. With the design itself $3.1\times$ worse without $S_{12,12}$ (above), two-qubit QNS is necessary both to *build* a decoupled entangler and to *certify* its error budget.
- **The prediction.** The predictions on this landscape are uniformly conservative ($\sim 1.5\times$ above truth: the reconstruction reads the quiet band slightly high and treats the flagged DC points pessimistically), and *certified*: every uncertainty band in Sec. 4 exists only because the bars in Table 2 are unbiased with stated coverage. With biased central values the same bands would be confidently wrong.
- **The spectroscopy itself.** The raw arm's spectra are relatively corrupted by up to $\sim 2\times$ on the line heights that set every trap's depth and $\sim 9\times$ on the quiet ZZ teeth that set the ZZ floor; mitigation recovers the reference to 0.02σ.

What the cross-spectra are for. The off-diagonal spectra ($S_{1,2}$, $S_{1,12}$, $S_{2,12}$) barely move either ladder above — and that is a theorem, not a landscape accident: in every PTM element the cross-terms enter through $\pm$ filter combinations whose first-order contributions cancel pairwise in the *average* fidelity, surviving only at $O(\chi^2)$. Average fidelity is the wrong meter for them. Their meter is element-resolved: *which* coherence survives. The idle's defining task makes this concrete (Fig. 4): hold half of a Bell pair, $\Phi^+ = (|00\rangle + |11\rangle)/\sqrt{2}$, through the idle. The Bell coherences are exactly ZZ-immune ($|00\rangle$ and $|11\rangle$ share the Z_1Z_2 eigenvalue), so only $S_{1,1}, S_{2,2}$ and $\operatorname{Re} S_{1,2}$ enter: Φ^+ decays through the $S_{1,1} + S_{2,2} + 2\operatorname{Re} S_{1,2}$ combination under in-phase decoupling ($y_2 = +y_1$: simultaneous pulses, the hardware default) and through $S_{1,1} + S_{2,2} - 2\operatorname{Re} S_{1,2}$ under anti-phase decoupling ($y_2 = -y_1$: the same train with one qubit's idle bracketed by frame flips). With the carrier common-mode at $c_{\mathrm{QS}} = 0.95$, the bare physics is a $33\times$ split: in *free* evolution, where nothing suppresses the carrier, Ψ^+ outlives Φ^+ by close to the full $(1 + c)/(1 - c) = 39\times$ factor (1.81×10^{-3} vs 5.93×10^{-2} at 320τ) — a decoherence-free-subspace structure that no single-qubit measurement can reveal. Under informed decoupling the carrier is largely suppressed and the choice still pays: the two implementations of the *same pulse train* — identical single-qubit marginals, average fidelities equal to *seven digits* (computed ratio 1.000 at every gate time), indistinguishable to any single-qubit QNS campaign — store the pair $1.2\times$ to $3.3\times$ apart, the split growing with storage time as the carrier's share of the decoupled residual grows. Which implementation protects the pair is decided by the *sign* of the measured low-frequency $\operatorname{Re} S_{1,2}$ (here > 0, so anti-phase wins; on a device with anti-correlated drift the assignment flips); the magnitude, measured by the comb at 5σ-plus, prices the payoff. The blind reconstruction predicts the right parity ordering at every gate time, and a record/replay Monte Carlo through the synthesized noise reproduces the formula values to 0.1–1.3%. The blind average-fidelity NT winner sits between the two parities — correctly, since average fidelity cannot see the choice. This is the cross-spectra story in one figure: invisible to average fidelity, decisive for entanglement storage, and only two-qubit QNS can measure it.

4 The gates: margins for both operations

Figure 5 summarizes the gate results on the 128k reconstruction, all under the 8τ pulse-spacing scenario and the blind protocol: select on the characterized spectra, report the true infidelity. Both June 12 requirements hold at once. *Realistic infidelities:* the tailored entangling gate holds in the 10^{-4}–10^{-3} band across $T_G = 40$–2560τ — 7.3×10^{-4} at 320τ, rising to 6.2×10^{-3} at 2560τ as the forced ZZ exposure accumulates — while tailored idling holds at the low-10^{-3} level out to $10240\tau \approx 3\,T_2^*$. *A robust margin over the best library sequence:* the entangling margin climbs from $1.5\times$ at 40τ to a $\sim 10\times$ plateau across 320–2560τ (9.8, 11.4, 9.7, $9.4\times$); the plateau — rather than the unbounded growth of a ZZ-free landscape — is itself the signature of the forced ZZ residual no CZ design can dodge. The idle margin grows from $\sim 2\times$ at short times to $\sim 24\times$ at 10240τ, for the structural reason

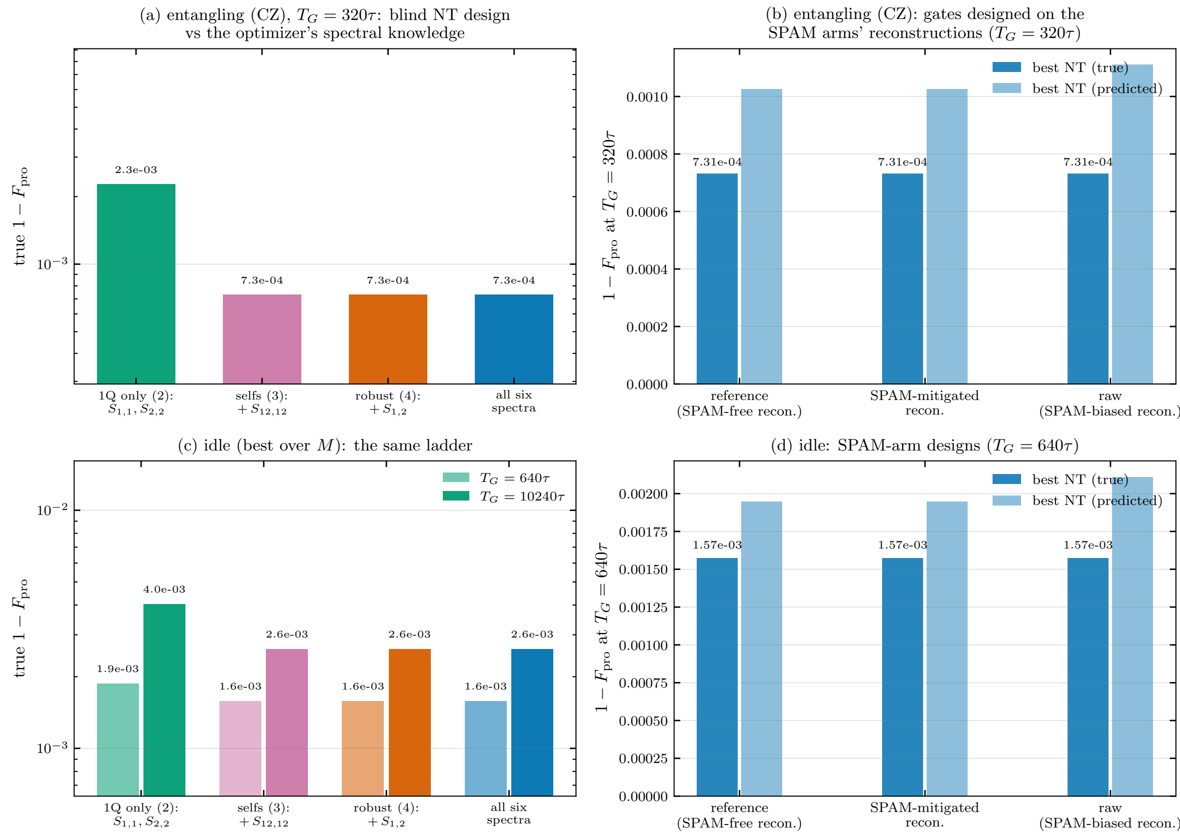

Figure 3: Both experiments for both gates: entangling (top row) and idle (bottom row, best over M; light/dark bars: $T_G = 640\tau$ and 10240τ). Left column: true infidelity of the blind NT winner as the optimizer's spectral knowledge grows from single-qubit-only, through the three selfs and the robust protocol's four, to all six spectra. The CZ ladder is flat — its trains barely touch the two-qubit structure; the idle ladder is not — the coupler line is invisible to a single-qubit campaign, and the designs that cannot see it pay. Right column: gates designed on the SPAM arms' reconstructions; true infidelity (solid) and the designer's own prediction (light). The raw arm's bias surfaces in the prediction, not the design.

of Sec. 1: each doubling of the gate time slides the CDD library's passbands onto the next defect-line harmonic, while the tailored design keeps using the measured quiet band.

The margins, with their 95% confidence intervals under reconstruction uncertainty:

- **Entangling (CZ):** 1.5× [1.79, 1.81] at 40τ; 2.2× [1.67, 1.71] at 80τ; 5.9× [4.14, 4.20] at 160τ; 9.8× [6.78, 6.92] at 320τ; 11.4× [9.45, 9.61] at 640τ; 9.7× [9.46, 9.57] at 1280τ; and 9.4× [9.56, 9.67] at $2560\tau \approx 0.8\,T_2^*$ — a plateau, not the unbounded climb of a ZZ-free landscape: the forced ZZ residual caps the tailored gate at depth as well.
- **Idle:** 2.1× [2.37, 2.53] at 320τ; 2.2× [2.12, 2.28] at 640τ; 2.9× [2.73, 3.01] at 1280τ; 2.4× [1.97, 2.31] at 2560τ; 14.2× [15.5, 20.0] at 5120τ; and 24.0× [24.8, 37.9] at $10240\tau \approx 3\,T_2^*$, once the deep idle's repetition count lets it dodge the ZZ knee.

In each entry the central value is the *true* margin and the bracket is the 95% interval of the *predicted* margin under reconstruction uncertainty. From 160τ on, the bands track the true margins and sit conservatively below them: the reconstruction's slightly-above-truth reading of the quiet band makes every predicted NT infidelity an upper estimate — the pipeline does not flatter its own gates. At the two shortest gate times, where best CDD and best NT sit within 2–3× of each other at the few-10^{-4} floor, the predicted ratio is least accurate (true $1.5\times/2.2\times$ vs. predicted $\sim 1.8\times$); the quoted true values are the honest ones there.

Where this leaves the two-scenario story: the anchored model says that on a quiet, monotonic, well-behaved

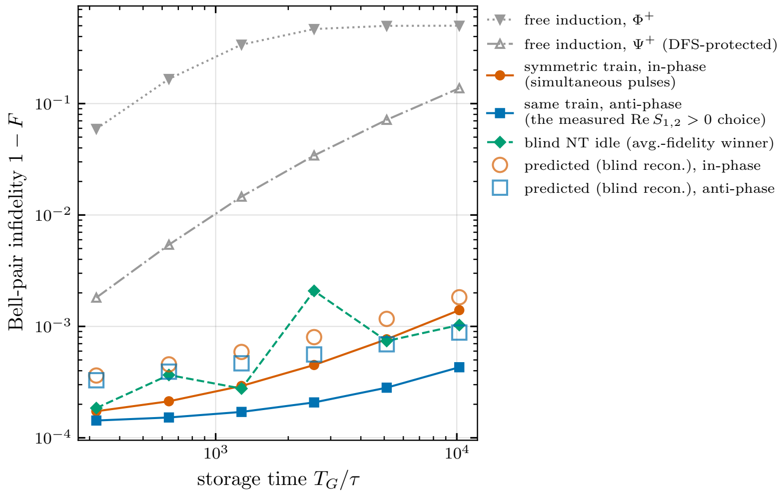

Figure 4: Holding a Bell pair through the idle. Free induction realizes the bare $(1 + c)/(1 - c) \approx 39\times$ ($33\times$ realized) split between Φ^+ (grey, filled) and Ψ^+ (grey, open): correlated-noise physics invisible to single-qubit data. Under decoupling, the same symmetric train implemented in-phase (simultaneous pulses, orange) and anti-phase (frame-flipped, blue) — implementations with identical average fidelity — still split by up to $3.3\times$, growing with storage time; open markers are the values predicted from the blind reconstruction, which picks the right parity everywhere. The blind average-fidelity NT idle (green) sits between the parities, as it must.

operation	FID	best CDD	best NT	margin
idling 640τ	1.67e−1	3.48e−3	**1.57**e−**3**	2.2×
idling 10240τ	5.28e−1	6.28e−2	**2.62**e−**3**	24.0×
entangling 320τ	6.12e−2	7.16e−3	**7.31**e−**4**	9.8×

Table 3: Sweet-spot numbers on the featured model (true process infidelities, blind designs, 8τ pulse-spacing scenario).

device, generic DD is nearly optimal and tailoring buys a reproducible 1.5–1.7×. The featured model says that the moment a device carries resolvable structure — defect harmonics, coupler resonances, common-mode drift, the things purification demonstrably does not remove — the same characterization pipeline, run once at a budget a real experiment can afford, separates 10^{-2}-class from 10^{-4}–10^{-3}-class two-qubit gates; that the parts of the reconstruction the average fidelity does not consume are exactly the parts that decide channel-resolved budgets and entanglement storage; and that no shortcut (a single-qubit campaign, unmitigated SPAM) delivers the same gates with the same certified predictions. Which device one has is an empirical question — and answering it is itself the reconstruction.

All optimization blind; true infidelities evaluated exactly on the analytic model; characterization at 128k shots per estimator, SPAM arms at 64k on identical noise; pulse budgets separation-limited under the 8τ spacing scenario for both the NT search and the CDD library.

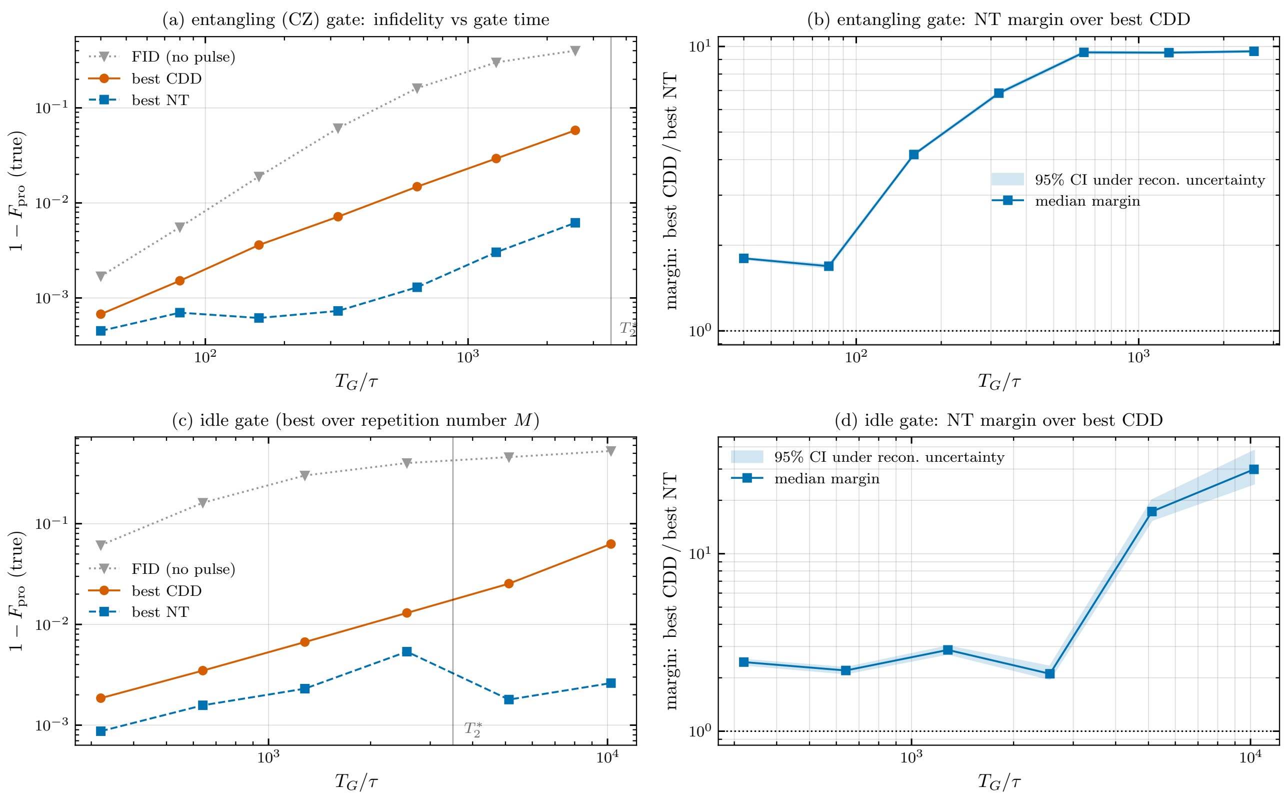

Figure 5: Noise-tailored gates on the featured model, all design blind, all scoring on truth, pulse-spacing constraint 8τ throughout. (a,c): true process infidelity vs. gate time for the entangling and idle gates: free induction (grey), best CDD (blind-selected, orange), best NT (blind-optimized, blue); the vertical line marks $T_2^* = 3500\tau$ (the idle is optimized over the repetition number $M \leq 128$ at each gate time). (b,d): the corresponding NT-over-CDD margin with the 95% band obtained by propagating the reconstruction's quoted statistical and systematic uncertainties (systematics drawn coherently per channel, tail fits redone per draw) through both fixed winner sequences; the band is on the *ratio*, so common-spectrum fluctuations cancel.